Original 8-bit RTL: High-Contrast Saturation Verification

VCS + Verdi nWave | All-00 Reference vs. All-FF Search Stress Fixture (520 MHz)

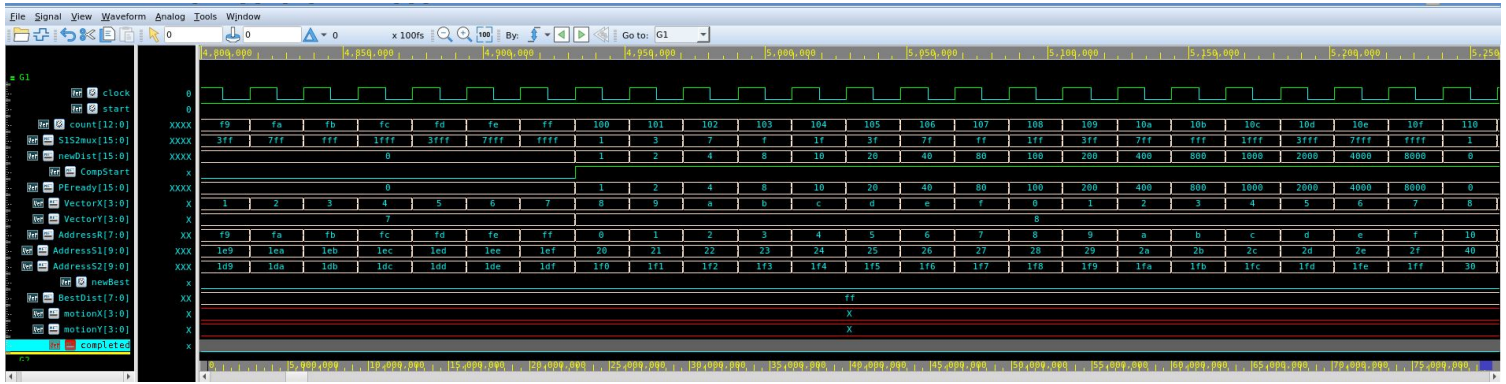


Figure 7: First row completion (count = 0x00F9..0x0110) — PEready[15:0] shifts 0x0001..0x8000, newBest stays 0, BestDist[7:0] = 0xFF

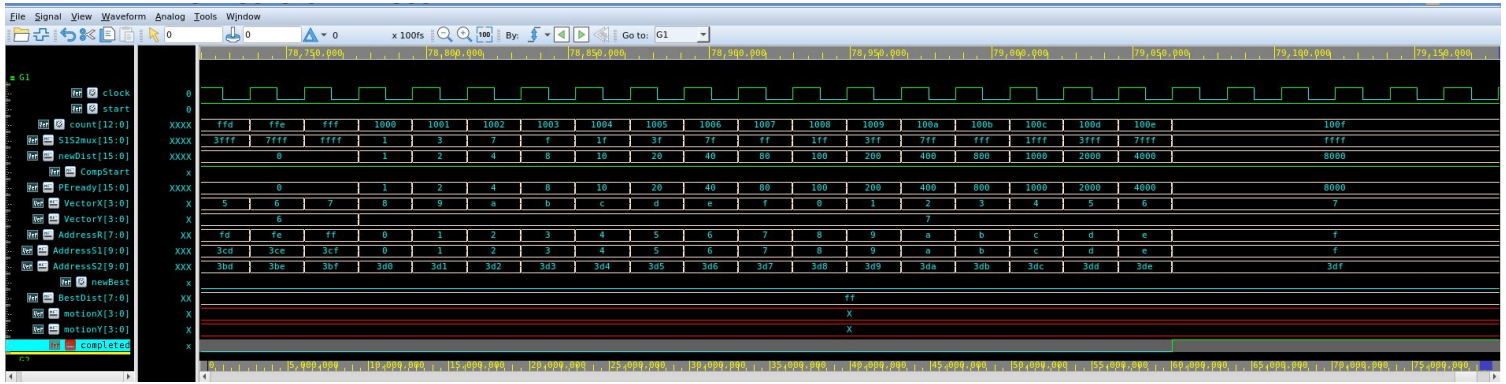


Figure 8: Module completion (count = 0x0FFD..0x100F, t = 7,905.45 ns) — completed = 1, BestDist[7:0] = 0xFF, motionX/Y = X (no candidate < 0xFF)

Corrected 16-bit SAD RTL: Full-Precision & Stimulus Comparison

VCS + Verdi nWave | 16-bit Accumulate/PEout/BestDist + Completion Hold at count = 13'h100F



Figure 9: Direct perfect-match fixture (ref.txt / search.txt) — BestDist[15:0] latches 0x0100 at count = 0x0100 and reaches 0x0000 at count =

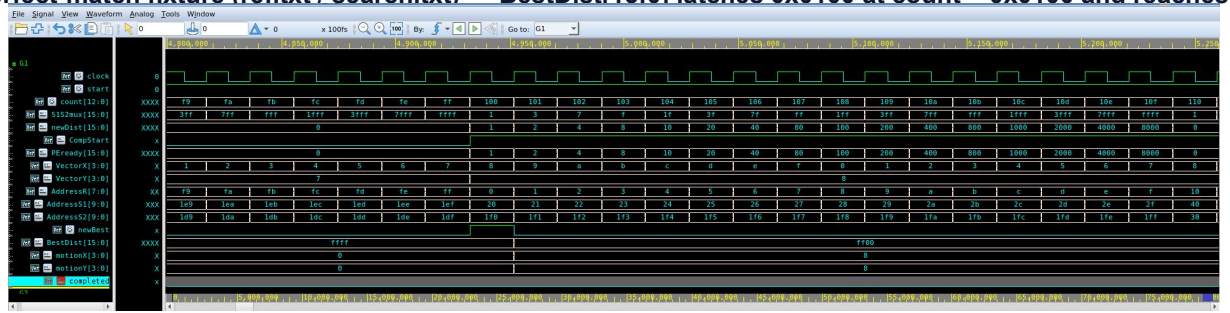


Figure 10: High-contrast stress fixture (All-00 vs. All-FF) — newBest pulses 1 at count = 0x0100, latching BestDist[15:0] = 0xFF00 (65,280) and vector (8,8)

Stimulus Comparison at 520 MHz: (1) Team / Direct Perfect Match -> BestDist = 0000 [motionX/Y = E,D or 7,8] | (2) Perturbed Match (+3) -> BestDist = 0003 [motionX/Y = E,D at count = 1542] | (3) All-00 vs. All-FF Stress -> BestDist = FF00 [motionX/Y = 8,8 at count = 256]

14 nm Physical Design & 3-Corner PrimeTime Sign-Off

Synopsys Fusion Compiler + IC Validator + PrimeTime | SAED14 RVT | 520 MHz (1.923 ns)

Physical Implementation & Sign-Off Metrics (V9):

- Die / Core Size: 125.07 x 62.80 um (7,854.40 um2 die; 6,558.99 um2 core)
- Standard-Cell Area: 3,416.18 um2 netlist (52.08% util; 6,221.28 um2 w/ fillers)
- Placed Cell Count: 8,230 logic cells (449 FFs); 15,983 total DEF components
- Synthesized Memories: 0 macros (ROM_R & ROM_S in testbench only)
- Total Power (TT 0.80V 25C): 0.890 mW (0.889 mW dynamic + 0.00089 mW leak)
- IC Validator DRC & LVS: CLEAN — 0 DRC violations (242/242 rules), 0 LVS errors

Standalone 3-Corner PrimeTime STA (SPEF Annotated):

- **FAST (FF 0.88V 125C / minTLU): Setup WNS = +0.71 ns (0 viol) | Hold WNS = +0.00 ns (0 viol)**
- **TYPICAL (TT 0.80V 25C / nomTLU): Setup WNS = +0.54 ns (0 viol) | Hold WNS = +0.04 ns (0 viol)**
- **SLOW (SS 0.72V -40C / maxTLU): Setup WNS = +0.22 ns (0 viol) | Hold WNS = +0.14 ns (0 viol)**

SDC Verified: 1.923 ns period (520 MHz), 0.500 ns I/O delay, 0.200 ns uncertainty.

GDSII Stream Out: 80 MB merged top.gds (0 unplaced cells, 0 GDS-110 warnings)

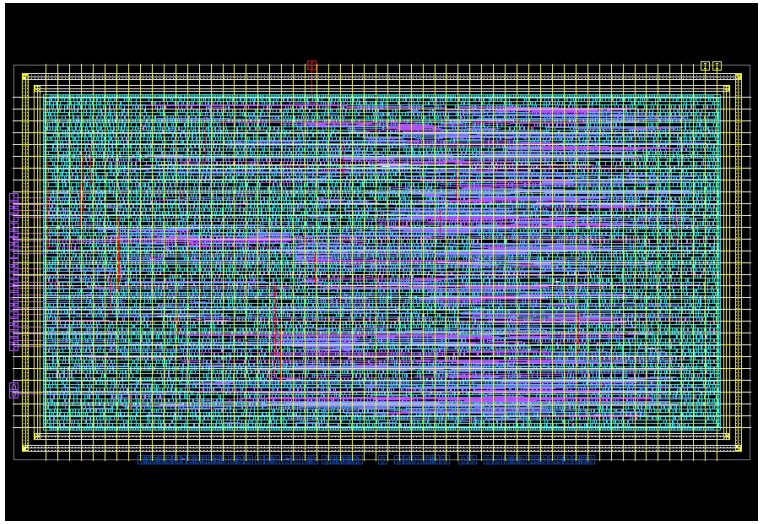


Figure 11: Final Routed 14 nm Chip Layout (top/v9_signoff_clean.design, 0 DRC, 3-Corner PT Closed)